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COMMENTARY

The epistemological trap from the real-world evidence concept

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Abstract

Objectives: 'Real-world' evidence (RWE) and related terms are gaining popularity in the scientific literature, usually in reference to evidence collected outside of traditional randomized controlled trials (RCTs). However, the concept lacks a standardized definition and rests on a poor epistemological foundation. This commentary explores the 'illusion of the real,' arguing that RWE-related terminology undermines scientific transparency by implying that routinely collected data capture an objective truth more effectively than controlled experiments.

Methods: A commentary with critical epistemological analysis of RWE.

Results: RWE underpins the epistemological illusion that the collection of observational or routine data is a passive, neutral, or intervention-free process. In fact, collecting data is, in itself, an active intervention by the researcher: an intervention that modifies, filters, and constructs the observed reality. This intervention introduces structural biases that remove any evidence of the supposed 'real truth.' Neither RCTs nor observational studies can perhaps access the absolute truth, since the act of observing and control already alters the phenomenon. The pursuit of 'real' evidence creates an anarchic methodological space where the lack of control is mistaken for authenticity, ultimately producing contextual constructions rather than objective truths.

Conclusion: Although data collected outside traditional RCTs are essential to decision-making, the 'real-world' label is a rhetorical construct that obscures study design and introduces an illusion of objective truth. Because all scientific inquiries are limited by the observer's methods, 'real-world' claims are essentially fallacious. To promote transparency, researchers should abandon RWE-related terminology in favor of precise study labeling and explicit disclosures of data source limitations. © 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Keywords: Real-world evidence; Epidemiology; Research design; Internal validity; Scientific transparency

1. Commentary

1.1. Evidence in or from the real world?

Real-world evidence (RWE) is not a standardized concept. RWE usually refers to evidence collected outside of traditional randomized controlled trials (RCTs) or, more broadly, data collected during routine health-care provision [1–4]. The applicability of this concept stems from some authors' criticism of the potential inability of formal methodological strategies to address the complexity of the real life [1].

The use of the term RWE is related to several problems and inconsistencies. "Real life" and "real people" are variable and complex terms by deduction contrasted with the 'artificial controlled life' or (or the scenario of the study) and 'artificial people' (or the participants of the study), as if these last belonged to a different and separate entity from the former. It is frequently challenging to define what 'real' (and 'artificial', by consequence) means with precision or to achieve consensus on their exact meaning [1]. The misuse of this terminology undermines the transparency of research and is not in line with the key reporting guidelines (eg, Consolidated Standards of Reporting Trials [CONSORT] and Strengthening the Reporting of Observational Studies in Epidemiology [STROBE]) [2].

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What is new?

Key findings

- 'Real-world evidence' falsely implies data collection is a neutral process.
- Scientific inquiry is an active intervention that constructs 'reality' and control bias.

What this adds to what was known?

- A critical epistemological analysis of the 'real-world' concept.

What is the implication, what should change now?

- Abandon vague 'real-world' labels for precise epidemiological terminology.
- Researchers must disclose data constraints instead of masking them as 'the real.'

This notion of 'real life' may also imply a hierarchical or even moralistic distinction between study designs, as if controlled experimental studies (the 'artificial life') provided less trustworthy or less generalizable evidence, while noncontrolled studies, by being allegedly closer to everyday clinical and dynamic reality (hence the term 'real'), would be more reliable and representative. Such a view inverts the traditional methodological paradigm of the past century, in which internal validity and control of bias were the main foundations of scientific trustworthiness.

Therefore, the use of this terminology neither clarifies nor adds information about the study design. Its use, because it is employed in various designs, is decreasing transparency and increasing the risk of misinterpretation of health research [2].

1.2. The purse of the real

The term RWE is identified in a wide range of publications, ranging from case series to clinical trials [2]. Some applications are quite similar, such as real-world studies or real-world data, while others extend further, referring to real-life evidence. Makady et al. addressed the absence of a unified definition for RWE by reviewing existing ones from the literature and stakeholders to identify commonalities and variations [3]. The authors identified 38 definitions for RWE. Twenty of them (53%) defined RWE as health data that are not collected in the setting of an RCT. Other categories included data from noninterventional/noncontrolled environments (24%), nonexperimental contexts (13%), and miscellaneous (11%) [3].

Although the definition of this term and its variations are not clear in the literature, the issue is not only centered on the heterogeneity of nomenclature but also on the underlying assumption that RCTs are incapable of capturing the real effect of the intervention. These terms have emerged in response to the desire from professionals to find a 'real effect', ie, one that is as close as possible to clinical practice, with all its natural and chaotic variables. This creates the false perception that avoiding the standardization of RCTs allows us to get closer to the truth.

The central problem with 'real-world' terms lies in the illusion that observational data capture a real or objective truth, when, in reality, the process of collection and analysis introduces inherent biases, subjectivity, and methodological limitations, making the final evidence constructed and fallible (far from real). This point is illustrated by a meta-epidemiological review, where Hemkens et al. found that analysis of routinely collected health data can produce divergent results from subsequent RCTs addressing identical clinical questions, often markedly overestimating treatment effects despite advanced techniques to adjust for confounding [5].

1.3. Is it possible to assess the real effect of an intervention?

Excessive control over the world to study it limits a study's external validity, as it modifies and restricts the natural origin of information. Conversely, not controlling the world (or controlling it less) compromises internal validity by substantially increasing the risk of bias in data generation and processing. This tension lies at the core of the methodological debate. Although formal scientific methodology has limitations inherent to the human condition and the nuances of evaluating complex psychobiological systems, its philosophical robustness allows it to produce information that is more aligned with objective reality.

There is a perception that moving away from more rigid methodological models gives us room to get closer to 'real life.' However, this proposal may fall into the fallacy that 'real evidence' is relative to collection methods, allowing for anarchic analyses, strictly selected populations, and ultimately generating biases. They end up inducing subjectivity, transforming data into contextual constructions.

However, what RWE appears to overlook is that subjectivity should enter at a different stage of scientific inquiry, namely, in the interpretation of results by clinicians. A scientific study does not see its findings immediately translated into medical practice upon publication. Critical appraisal of the results is required, followed by deliberate consideration of whether to generalize findings and apply them to particular settings. Requiring exact alignment between study populations or conditions and the intricate real-world contexts of application can unduly restrict the evidence's utility [1].

Moreover, the RWE underpins the epistemological illusion that the collection of observational or routine data is a passive, neutral, or intervention-free process. In fact, collecting data is, in itself, an active action by the researcher: an intervention that modifies, filters, and constructs the observed reality. This intervention introduces structural biases that remove any evidence of the supposed 'real truth'. Neither RCTs nor observational studies can perhaps access the absolute truth, since the act of observing and control already alters the phenomenon. However, the systematization of these methods allows us intellectual honesty.

Compliance with systematized methodological precepts allows us to offer transparency in the search for evidence. This is because, invariably, we have an obstacle to overcome: the dissociation, recognized by modern philosophers, between what we perceive (structured by the mind's a priori categories: space, time, and causality), called by Immanuel Kant the phenomenon (*Phänomen*), and the noumenon (the 'thing in itself', *Ding an sich*), which is directly inaccessible [6].

Edmund Husserl founded phenomenology as a philosophical method aiming to return 'to the things themselves', ie, to experience as it appears to consciousness before any theoretical interpretation. He proposed that all knowledge begins with lived experience (e*rlebnis*), and that philosophy must describe these experiences rigorously by suspending assumptions about the external world, a process known as *epoché*. At its core, Husserl's thought rests on an idea that consciousness and world are always correlated: every act of consciousness is *consciousness of something* (*intentionality*). Knowledge, therefore, is not a passive reflection of reality but a structured relationship between subject and phenomenon [7].

Werner Heisenberg pointed out that '*What we observe is not nature in itself but nature exposed to our method of questioning'* [8]. The act of measuring disturbs the observed system, not because of technical failure but because of the ontological limitation of tools of measuring the reality. Therefore, proper methodological compliance honestly allows this data to be submitted to the scrutiny of the scientific community.

So, is it possible to assess the real effects of interventions? The answer is not at all. As described above, there are limitations inherent to the scientific method and practice. It is preferable to measure phenomena under controlled conditions and to be fully aware of the method's limitations, rather than to measure them without control, thereby introducing bias while believing one is getting closer to the truth. We must use transparency to promote scientific advancement in the most appropriate way possible.

Even if the term 'real' in RWE denotes studies conducted outside typical clinical research settings, this distinction is entirely compatible with the principles of planned interventions and randomization [4]. Thus, terms that perpetuate the fallacy of unmediated 'real' should be abandoned in favor of a critical epistemology that rigorously assesses the quality, provenance, and inherent limitations of evidence. Rather than invoking the rhetorical allure of the 'real,' our field must adopt a transparent, humble, and methodologically reflexive stance.

It is better to accept the limitations of a controlled setting—and thereby maintain awareness and management of bias—than to pursue greater external validity at the cost of introducing uncontrolled bias and loss of transparency.

1.4. Avoiding the trap

This commentary discusses the illusion created by the term RWE, which implies that there is a distinct kind of data that reflect the truth of the 'real world.' Data collected outside traditional RCTs are essential to decision-making, and methods for their collection, appraisal, and use have been developed and discussed in clinical epidemiology long before the term RWE was introduced.

To avoid falling into this trap, it is essential to discard the use of RWE-related terminology. Epidemiological studies should be identified by their design (eg, cohort, case-control, and cross-sectional), and all data sources should be clearly stated. Moreover, it is imperative to underscore the inherent structural constraints and biases of each data source, rather than obscuring them with a vague and poorly defined term.

CRediT authorship contribution statement

Leonardo Y. Kasputis Zanini: Writing − original draft, Conceptualization. **Diego Adão:** Writing − review & editing, Supervision, Investigation, Conceptualization. **Gabriela CL. Martins:** Writing − original draft, Conceptualization. **Rachel Riera:** Writing − review & editing, Validation, Supervision, Investigation, Conceptualization. **Rafael Leite Pacheco:** Writing − review & editing, Supervision, Project administration, Investigation, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

Data availability

No data were used for the research described in the article.

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